

Exposé II. Topologies and Sheaves

Opening; Sections 1–2.7

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After the definition of topologies and pretopologies (no. 1) and of sheaves of sets (no. 2), no. 3 turns to the central theorem of this exposé (3.4): the existence of the sheaf associated to a presheaf. In order to cover all cases encountered in practice, the existence of this functor is proved for presheaves on a $\mathcal{U}$ -site (3.0.2). Nos. 4 and 5 draw the consequences of this theorem for the behavior of inductive and projective limits in categories of sheaves. In no. 6 one defines and studies sheaves with values in arbitrary categories, and then immediately turns attention to sheaves of rings, groups, modules, etc.

1 Topologies, covering families, pretopologies

Definition 1.1. A *topology* on a category C consists, for each object X of C , of a set $J(X)$ of sieves on X , subject to the following axioms:

- T 1)** *Stability under base change.* For every object X of C , every sieve $R \in J(X)$, and every morphism $f : Y \rightarrow X$ with $Y \in \text{Ob}(C)$, the sieve $R \times_X Y$ on Y belongs to $J(Y)$.
- T 2)** *Local character.* If R and R' are two sieves on X , if $R \in J(X)$, and if for every $Y \in \text{Ob}(C)$ and every morphism $Y \rightarrow R$ the sieve $R' \times_X Y$ belongs to $J(Y)$, then R' belongs to $J(X)$.
- T 3)** For every object X of C , the maximal sieve X belongs to $J(X)$.

1.1.1

The sieves belonging to $J(X)$ will be called the *sieves covering* X , or also the *refinements* of X . From axioms (T1), (T2), and (T3), one immediately deduces that the set of sieves covering X is stable under finite intersections, and that every sieve containing a covering sieve is again a covering sieve. Thus $J(X)$, ordered by inclusion, is cofiltered (I 8).

1.1.2

Let C be a category, and let T and T' be two topologies on C . The topology T is said to be *finer than* the topology T' if, for every object X of C , every refinement of X for the topology T' is a refinement of X for the topology T . In this way one defines an order structure on the set of topologies.

1.1.3

Let $(T_i)_{i \in I}$ be a family of topologies on C . For each object X of C , the sets of sieves on X which are covering for all the topologies T_i satisfy axioms (T1), (T2), and (T3), and therefore define a topology: the *intersection topology* of the T_i , i.e. the greatest lower bound of the T_i . It is the finest of the topologies which is less fine than all the topologies T_i . It follows that the family $(T_i)_{i \in I}$ has a *least upper bound*: the intersection topology of the topologies finer than each of the T_i .

1.1.4

In particular, the assignment $J(X)$ = the set of all sieves on X is a topology finer than every topology on C ; we shall call it the *discrete topology*.

There is a topology less fine than every topology on C : the topology for which $J(X) = \{X\}$ for every X of C . This topology is called the *coarse* or *chaotic* topology.

1.1.5

A category C endowed with a topology is called a *site*. The category C is called the *underlying category* of the site.

Definition 1.2. Let C be a site and let X be an object of C . A family of morphisms $(f_\alpha : X_\alpha \rightarrow X)$, $\alpha \in A$, is called *covering* if the sieve generated by the family f_α (I 4.3.3) is a sieve covering X .

1.1.6

Let C be a category. Suppose that for every object X of C we are given a set of families of morphisms with target X . Then there exists a topology T which is the least fine among the topologies for which the given families are covering, namely the intersection (1.1.3) of all topologies in question. We call this topology the *topology generated by the given sets of families of morphisms*. In general it is awkward to describe all covering sieves of this topology. The situation is more agreeable in the following case.

Definition 1.3. Let C be a category. A *pretopology* on C consists, for each object X of C , of a set $\text{Cov}(X)$ of families of morphisms with target X , subject to the following axioms:

PT 0)

For every object X of C , the morphisms occurring in the families of morphisms of $\text{Cov}(X)$ are base-changeable. Recall that a morphism $Y \rightarrow X$ is called base-changeable if for every morphism $Z \rightarrow X$, the fiber product $Z \times_X Y$ is representable.

PT 1)

For every object X of C , every family $(X_\alpha \rightarrow X)_{\alpha \in A}$ belonging to $\text{Cov}(X)$, and every morphism $Y \rightarrow X$ with $Y \in \text{Ob}(C)$, the family

$$(X_\alpha \times_X Y \rightarrow Y)_{\alpha \in A}$$

belongs to $\text{Cov}(Y)$. This is stability under base change.

PT 2)

If $(X_\alpha \rightarrow X)_{\alpha \in A}$ belongs to $\text{Cov}(X)$, and if for each $\alpha \in A$ the family

$$(X_{\beta_\alpha} \rightarrow X_\alpha)_{\beta_\alpha \in B_\alpha}$$

belongs to $\text{Cov}(X_\alpha)$, then the family $(X_\gamma \rightarrow X)_{\gamma \in \coprod_{\alpha \in A} B_\alpha}$, where for $\gamma = (\alpha, \beta_\alpha)$ the morphism $X_\gamma \rightarrow X$ is the composite

$$X_\gamma = X_{\beta_\alpha} \rightarrow X_\alpha \rightarrow X,$$

belongs to $\text{Cov}(X)$. This is stability under composition.

PT 3)

The family $X \xrightarrow{\text{id}_X} X$ belongs to $\text{Cov}(X)$.

1.3.1

Definition 1.1.6 allows us, for a given *pretopology* on C , to consider the *topology* on C generated by this pretopology. Note that if C is a category in which fiber products are representable, then every topology T on C can be defined by a pretopology, namely the one for which $\text{Cov}(X)$ consists of *all* families covering X for the topology T .

Proposition 1.4. Let C be a category, let E be a pretopology on C , let T be the topology defined by the pretopology E (1.1.6), and let X be an object of C . Denote by $J_E(X)$ the set of sieves on X generated by the families of the pretopology, and by $J_T(X)$ the set of refinements of X for the topology T . Then $J_E(X)$ is cofinal in $J_T(X)$. In other words, for a sieve R on X to belong to $J_T(X)$, it is necessary and sufficient that there exist a sieve $R' \in J_E(X)$ such that $R' \subset R$.

Proof. For each object X of C , let $J'(X)$ be the set of sieves on X which contain a sieve of $J_E(X)$. Evidently $J'(X) \subset J_T(X)$. To show that $J'(X) = J_T(X)$, it is enough to show that the data of the $J'(X)$ define a topology on C . Now the $J'(X)$ plainly satisfy axioms (T1) and (T3). It remains to verify (T2). For this it is enough to prove that, if R' is a subsieve of $R \in J_E(X)$ such that for every $Y \rightarrow R$ the sieve $R' \times_X Y$ on Y belongs to $J'(Y)$, then R' belongs to $J'(X)$. The sieve R is generated by a family $(X_\alpha \rightarrow X)$ belonging to $\text{Cov}(X)$, and for every α the sieve $R' \times_X X_\alpha$ contains a sieve generated by a family $(X_{\beta_\alpha} \rightarrow X_\alpha)$ belonging to $\text{Cov}(X_\alpha)$. It follows that R' contains a sieve generated by the family $(X_{\beta_\alpha} \rightarrow X)$. Hence, by axiom (PT2), R' contains a sieve of $J_E(X)$, and therefore belongs to $J'(X)$. $\square$

2 Sheaves of sets

Definition 2.1. Let C be a site whose underlying category is a $\mathcal{U}$ -category. A presheaf F with values in $\mathcal{U}\text{-Set}$ is called *separated* (respectively, a *sheaf*) if, for every sieve R covering an object X of C , the map

$$\text{Hom}_{\widehat{C}}(X, F) \longrightarrow \text{Hom}_{\widehat{C}}(R, F)$$

is injective (respectively, bijective). The full subcategory of $\widehat{C}$ whose objects are the sheaves is called the category of sheaves of sets on C , and is most often denoted $\widetilde{C}$. When no ambiguity can result, we shall simply say the category of sheaves on C ; conversely, we shall sometimes be more precise and say the category of sheaves with values in $\mathcal{U}\text{-Set}$, or the category of $\mathcal{U}$ -sheaves.

Proposition 2.2. Let C be a $\mathcal{U}$ -category, and let $\mathcal{F} = (F_i)_{i \in I}$ be a family of $\mathcal{U}$ -presheaves on C . For each object X of C , denote by $J_{\mathcal{F}}(X)$ the set of sieves $R \rightarrow X$ such that, for every morphism $Y \rightarrow X$ of C with target X , the sieve $R \times_X Y$ has the following property: the map

$$\text{Hom}_{\widehat{C}}(Y, F_i) \longrightarrow \text{Hom}_{\widehat{C}}(R \times_X Y, F_i)$$

is bijective (respectively, injective) for every $i \in I$. Then the sets $J_{\mathcal{F}}(X)$ define a topology on C , which is the finest topology for which each of the F_i is a sheaf (respectively, a separated presheaf).

Proof. The $J_{\mathcal{F}}(X)$ plainly satisfy axioms (T1) and (T3). It remains to show that they satisfy (T2). For this it is enough to show that:

- 1) if $R' \hookrightarrow R \hookrightarrow X$ are two sieves on X , with $R \in J_{\mathcal{F}}(X)$, and such that for every $Y \rightarrow R$, where Y is an object of C , the sieve $R' \times_X Y$ belongs to $J_{\mathcal{F}}(Y)$, then R' belongs to $J_{\mathcal{F}}(X)$;
- 2) if $R' \hookrightarrow R \hookrightarrow X$ are two sieves on X such that $R' \in J_{\mathcal{F}}(X)$, then R belongs to $J_{\mathcal{F}}(X)$.

In case 2), note that for every $Y \rightarrow R$, where Y is an object of C , the sieve $R' \times_X Y$ belongs to $J_{\mathcal{F}}(Y)$. In cases 1) and 2), the sieve R is the inductive limit of the objects Y of C lying over R (I 3.4). Inductive limits in $\widehat{C}$ are universal (I 3.3). Consequently, in cases 1) and 2), R' is the inductive limit of the $R' \times_X Y$. But in cases 1) and 2), $R' \times_X Y$ belongs to $J_{\mathcal{F}}(Y)$. Passing to the inductive limit over the objects Y of C lying over R , one deduces that the map

$$\mathrm{Hom}_{\widehat{C}}(R, F_i) \longrightarrow \mathrm{Hom}_{\widehat{C}}(R', F_i)$$

is bijective (respectively, injective) for every $i \in I$. It follows that the maps

$$\mathrm{Hom}_{\widehat{C}}(X, F_i) \longrightarrow \mathrm{Hom}_{\widehat{C}}(R', F_i), \quad \mathrm{Hom}_{\widehat{C}}(X, F_i) \longrightarrow \mathrm{Hom}_{\widehat{C}}(R, F_i)$$

are, in cases 1) and 2), bijections (respectively, injections). Moreover, hypotheses 1) and 2) are visibly stable under arbitrary base change $Y \rightarrow X$, with Y an object of C . Hence, for every object Y of C lying over X , and every $i \in I$, the maps

$$\begin{aligned} \mathrm{Hom}_{\widehat{C}}(Y, F_i) &\longrightarrow \mathrm{Hom}_{\widehat{C}}(R \times_X Y, F_i), \\ \mathrm{Hom}_{\widehat{C}}(Y, F_i) &\longrightarrow \mathrm{Hom}_{\widehat{C}}(R' \times_X Y, F_i) \end{aligned}$$

are in both cases bijections (respectively, injections). This shows that R' and R belong to $J_{\mathcal{F}}(X)$. $\square$

Corollary 2.3. Let C be a $\mathcal{U}$ -category, and for every object X of C let $K(X)$ be a set of sieves on X . Assume that the $K(X)$ satisfy axiom (T1) of (1.1). For a presheaf F to be a sheaf (respectively, a separated presheaf) for the topology generated (1.1.6) by the $K(X)$, it is necessary and sufficient that, for every object X of C and every sieve $R \in K(X)$, the map

$$\mathrm{Hom}_{\widehat{C}}(X, F) \longrightarrow \mathrm{Hom}_{\widehat{C}}(R, F)$$

be bijective (respectively, injective).

Corollary 2.4. In particular, let C be a $\mathcal{U}$ -category endowed with a pretopology. For a presheaf F to be a sheaf (respectively, a separated presheaf), it is necessary and sufficient that, for every object X of C and every family $(X_\alpha \rightarrow X)$ belonging to $\mathrm{Cov}(X)$, the diagram of sets

$$F(X) \longrightarrow \prod_{\alpha \in A} F(X_\alpha) \rightrightarrows \prod_{(\alpha, \beta) \in A \times A} F(X_\alpha \times_X X_\beta)$$

be exact (respectively, that the map

$$F(X) \longrightarrow \prod_{\alpha \in A} F(X_\alpha)$$

be injective).

Proof. Apply 2.3, then I 3.5 and I 2.12.

With Corollary 2.4 one recovers the definition given in [1].

Definition 2.5. Let C be a $\mathcal{U}$ -category. The *canonical topology* of C is the finest topology for which the representable functors are sheaves (2.2). A sieve covering X for the canonical topology will be called a *universally strict epimorphic sieve*. A covering family for the canonical topology will be called a *universally strict epimorphic family*. When, moreover, the morphisms of the covering family are base-changeable, the family will be called *universally effective epimorphic* [2].

Remark 2.5.1. For almost all the sites that have had to be used up to now, the topology is *less fine* than the canonical topology; in other words, the representable functors on C are sheaves, i.e. the covering families of C are universally strict epimorphic. The only exception to this rule is the site $\widehat{C}$ studied in no. 5, whose topology is *finer*, and most often strictly finer, than the canonical topology.

Proposition 2.6. For a sieve $R \hookrightarrow X$ to be universally strict epimorphic, it is necessary and sufficient that, for every object $Y \rightarrow X$ of C lying over X , and for every object Z of C , the map

$$\mathrm{Hom}_C(Y, Z) \longrightarrow \varprojlim_{C/(Y \times_X R)} \mathrm{Hom}_C(\cdot, Z)$$

be a bijection.

Proof. Immediate by applying 2.2 and then I 5.3. □

Remarks 2.7.

- 1) Proposition 2.6 gives a characterization of universally strict epimorphic sieves of a category C which is independent of the universe in which the presheaves take their values, under the sole condition that the Hom-sets $\mathrm{Hom}(X, Y)$ of C belong to this universe. Thus it allows one to define the canonical topology for any category C , without needing to specify the universes.
- 2) Let C be a site whose underlying category is a $\mathcal{U}$ -category, let F be a $\mathcal{U}$ -presheaf, and let $\mathcal{V}$ be a universe containing $\mathcal{U}$. The underlying category of C is a $\mathcal{V}$ -category, and F may be regarded as a $\mathcal{V}$ -presheaf. The $\mathcal{U}$ -presheaf F is a sheaf (respectively, a separated presheaf) if and only if the $\mathcal{V}$ -presheaf F is a sheaf (respectively, a separated presheaf). In other words, the condition for a presheaf F to be a sheaf (respectively, a separated presheaf) does not depend, in the sense just specified, on the universe in which the presheaf F takes its values. In particular, let C be a site and let F be a $\mathcal{U}$ -presheaf; one says that F is a sheaf (respectively, a separated presheaf) if there exists a universe $\mathcal{V}$ containing $\mathcal{U}$ such that the category C is a $\mathcal{V}$ -category and such that F is a $\mathcal{V}$ -sheaf (respectively, a separated $\mathcal{V}$ -presheaf). This property does not depend on the universe $\mathcal{V}$.